

Math 55 worksheet Monday 3/2/2026

1. Determine which amounts of postage can be formed using three-cent and five-cent stamps. (For example, eight cents can be formed; two cents cannot be.) Prove this using strong induction.
2. Let $\phi = \frac{1+\sqrt{5}}{2}$ and $\psi = \frac{1-\sqrt{5}}{2}$. Let f_n be the n -th Fibonacci number, starting $f_1 = 1, f_2 = 1, f_3 = 2$. Prove that $f_n = \frac{1}{\sqrt{5}}(\phi^n - \psi^n)$.
3. Once again, let f_n be the n -th Fibonacci number. Prove that $f_{n+1}f_{n-1} - f_n^2 = (-1)^n$ for all $n \geq 1$. (We say that $f_0 = 0$.)
4. Let g be a recursive function defined by $g(k) = 1 - g(k-1)$. Prove that, for any value of $g(0)$, there exists some M such that $|g(k)| < M$ for all k . That is to say, no matter where you start, g does not grow infinitely.